

# Muhammad Junayed

CUET, Raozan, Chattogram, Bangladesh | mdjunayed573@gmail.com | +880 1876220119 | LinkedIn | GitHub

## EDUCATION

### Bachelor of Science in Electronics and Telecommunication Engineering

March 2022–Present

Chittagong University of Engineering and Technology

- CGPA: 3.51/4.00 (upto 7th semester)

## WORK EXPERIENCES

### PORIDHI.IO

May 2025 – November 2025

Software Engineer Intern

- Worked on software engineering, cloud computing and backend development tasks during internship tenure.

### BRAIN STATION 23 PLC

May 2025

Industrial Trainee

- Completed industrial training focused on modern software engineering workflows and enterprise systems.

## RESEARCH & PUBLICATIONS

### B.Sc. Thesis:

- **Closing the Loop on RAG Hallucinations: Inference-Time Control via Dual Residual-Stream and FFN Activation Probes**

**Supervisor:** Priyonti Paul Tumpa, Assistant Professor, ETE, CUET

### Conference Publications:

- **Muhammad Junayed**, Tanzeem Tahmeed Reza, Md. Saiful Islam. **Enhancing Defect Recognition: Convolutional Neural Networks for Silicon Wafer Map Analysis**. In: 2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), Gazipur, Bangladesh. IEEE, 2024. DOI: 10.1109/ICAEEE62219.2024.10561853.
- **Muhammad Junayed**, Tanzeem Tahmeed Reza. **Leveraging Vision Transformer to Classify Cancer Stages from Breast Ultrasound Images**. In: IEEE International Conference on Signal Processing, Information, Communication and Systems 2025 (SPICSCON), Rajshahi University, Bangladesh, 2025. DOI: 10.1109/SPICSCON69221.2025.11504239.
- Md. Abdur Rahman, Md. Al Amin, Sabik Aftahee, **Muhammad Junayed**, Md. Ashiqur Rahman. **SmolLab\_SEU at BEA 2025 Shared Task: A Transformer-Based Framework for Multi-Track Pedagogical Evaluation of AI-Powered Tutors**. In: Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025). Association for Computational Linguistics (ACL), 2025, pp. 1127–1134.

## PROJECTS

### AroBot – Agentic RAG Multi-Modal Chatbot System

- Built a production-ready medical assistant using Llama 3.2:3B and LLaVA:7B via Ollama, with Pinecone vector search and FastAPI deployment.
- Designed modular agentic workflows for text, image, and PDF processing using OCR and RAG pipelines.
- Implemented prescription OCR, CLIP vision pipeline, LangSmith tracing, and DuckDuckGo search integration.
- Added conversation memory, rate-limited APIs, and a responsive chat UI so clinicians get auditable session histories, health metrics, and ingestion scripts for continuously updating vector indexes.

### Uber Fare Prediction with MLOps Orchestration

- Built an end-to-end MLOps pipeline using Airflow for scheduling, ZenML for pipeline orchestration, and Flask for deployment.
- Used MLflow for experiment tracking and selection of optimal ensemble models for production.
- Integrated PostgreSQL for metadata storage and system-level logging to support reproducibility and monitoring.

## Face Recognition & RFID Smart Attendance System

- Developed a Dockerized smart attendance system on Jetson Nano integrating MTCNN for face detection and ResNet-18 for face recognition, enabling real-time identity verification.
- Integrated JT308 RFID module and Canon EOS 250D camera for multi-modal attendance tracking, improving robustness and reducing spoofing in real-world deployment.

## Bangladesh Medicine Scrape

- Built a Django + Scrappy + Playwright pipeline that reuses verified browser cookies to crawl *MedEx* data.
- Stored results in PostgreSQL and exposed via Django Admin and token-authenticated REST APIs (DRF).

## NodeScope – Graph Algorithm Visualizer + Graph Type Predictor

- Built a full-stack React + Flask/TensorFlow platform for graph visualization and graph-type prediction using LightGIN GNN showing step-by-step BFS/DFS visualization and CI/CD pipeline.

## Credit Card Fraud Detection

- Built ML models using Random Forest, XGBoost, and LightGBM on highly imbalanced transaction datasets, achieved approximately 0.98 AUPRC and 0.96 F1-score with Streamlit deployment.

## Asynchronous Task Processing System (Flask + Celery + AWS Pulumi)

- Built distributed asynchronous task processing pipelines using Flask, Celery, RabbitMQ, Redis, Docker, and AWS EC2.
- Automated infrastructure provisioning using Pulumi (IaC) and implemented both single-node and multi-node EC2 deployment.

## RickshawX – Smart Mobility for CUET (Microservices Ride-Sharing Platform)

- Built a full-stack ride-sharing platform for CUET using a scalable **microservices architecture** with Node.js/Express, React, MongoDB, Redis, RabbitMQ, Docker, and Nginx.
- Developed independently deployable Auth, Ride, Trip, Payment, and Notification services with JWT authentication, centralized API Gateway routing, and asynchronous inter-service communication.
- Implemented an **event-driven system** using RabbitMQ, GPS-based fare calculation with the Haversine formula, Redis caching, trip management and real-time notification.

## TECHNICAL SKILLS

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- **Programming Languages:** Python, C++, PHP, Node.js, HTML, CSS, SQL
- **Frameworks & Tools:** Flask, FastAPI, React, Django, Docker, RabbitMQ, Redis, Pulumi, Numpy, Pandas, Matplotlib, Scikit learn, TensorFlow, Keras, OpenCV.
- **MLOps & Data Science:** AWS fundamentals, Container Networking, PostgreSQL, BigQuery, Power BI, MongoDB
- **Software:** MATLAB, VSCode, Cisco Packet Tracer, Adobe Photoshop, DaVinci Resolve
- **Typography:** LaTeX, MS Word, MS PowerPoint.

## EXTRACURRICULAR ACTIVITIES

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- **Televerse 1.0, CUET**, Team Lead & Organizer, of Bangladesh's first-ever Shark Tank-style hardware software integrated automation competition *IOTrix*. (Oct 2025–21 Nov 2025)
- **CUET Photographic Society (CUETPS)**, Assistant Design Secretary. (Sep 2024–11 Mar 2026)
- **3rd International Conference on Electrical, Computer and Communication Engineering (ECCE)**, CUET, Event Manager, Chattogram, Bangladesh. (Feb 2023)

## AWARDS AND CERTIFICATIONS

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- **Champion** – ETE Infixon Case Solving Competition, CUET
- **1st Runner-up** – Programming Hackathon, Department of ETE, CUET
- **Finalist** – Revealing The Unseen: Season VI by CUET Photographic Society-CUETPS
- **13th at Datathon**, (out of 108 teams), BitFest, Dept. of CSE, KUET, Bangladesh (2025)
- **Finalist** – SemEval 2024 Task 8: DataBench
- **3rd Place** – Techathon: The Heart of Robolution 1.0 by by the Department of Mechatronics and Industrial Engineering, CUET.
- **Finalist** – BEA 2025 Shared Task on Pedagogical Evaluation of AI Tutors, organized by ACL SIGEDU; secured Top 10 in Tutor Identity (F1: 0.8621) and competitive ranks in 4 additional tracks
- **Honorable Mention** – Break The Monolith Microservice Hackathon by IEEE Computer Society, CUET